Anomaly Detection and Classification in Multispectral Time Series Based on Hidden Markov Models

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***Abstract*— Monitoring agriculture from satellite remote sensing data, such as multispectral images, has become a powerful tool since it has demonstrated a great potential for providing timely and accurate knowledge of crops. Detecting anomalies in time series of multispectral remote sensing images for crop monitoring is generally performed using a large sample of historical data at a pixel level. Conversely, this article presents a framework for anomaly detection (AD), localization, and classification that exploits the temporal information contained in a given season at a parcel level to detect and localize outliers using hidden Markov models (HMMs). Specifically, the AD part is based on the learning of HMM parameters associated with unlabeled normal data that are used in a second step to detect abnormal crop parcels referred to as anomalies. The learned HMM can also be used in time segments to temporally localize the anomalies affecting the crop parcels. The detected and localized anomalies are finally classified using a supervised classifier, e.g., based on support vector machines. The proposed framework is applicable to images partially covered by clouds and can handle a set of crop parcels acquired in the same season bypassing problems due to crop rotations. Numerical experiments are conducted on synthetic and real data, where the real data correspond to vegetation indices extracted from several multitemporal Sentinel-2 images of rapeseed crops. The proposed approach is compared to standard AD methods yielding better detection rates with the advantage of allowing anomalies to be localized and characterized.**

***Index Terms*— Agricultural monitoring, anomaly classification, anomaly detection (AD), hidden Markov models (HMMs), remote sensing, time series.**

Manuscript received January 18, 2021; revised May 27, 2021; accepted July 13, 2021. Date of publication August 9, 2021; date of current version January 17, 2022. This work was supported in part by Minciencias through the 910-2019-Program ECOS Nord (Fusión de imágenes multitemporales ópticas y de radar de apertura sintética y su aplicación en detección de anomalías en cítricos) under Grant 8598 and in part by the Vicerrectoría de Investigación y Extensión of Universidad Industrial de Santander under Project 2730. *(Corresponding author: Jean-Yves Tourneret.)*

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Digital Object Identifier 10.1109/TGRS.2021.3101127

I. Introduction

MULTISPECTRAL images have been widely used in many studies to explore the vegetation properties of plants through the extraction of vegetation indices (VIs) [1]–[4]. In the last decade, researchers have proposed to use multitemporal images for several applications, including change detection [5] and land-cover classification [6], [7], where the challenge is mainly to exploit the redundancy and correlation across the spatial, spectral, and temporal dimensions of the images.

A. Hidden Markov Models for Vegetation Dynamics

Hidden Markov models (HMMs) are classical tools to analyze time series, allowing temporal correlations to be extracted with the introduction of latent variables interacting with the data [8]–[10]. Different works have shown that HMMs are valuable tools for modeling the dynamic behavior of crops across time, where the dynamics of vegetation is related to the phenology, chemical nutrients, climatic conditions, or water stress of crops [11], [12]. Some specific tasks for crop analysis based on HMM include crop recognition [13], crop classification [8], [14], and time evolution featuring [12]. In addition, an analysis of the normalized difference vegetation index (NDVI) using the HMM framework is proposed in [11], where the NDVI changes are used to characterize the dynamics of the vegetation during a temporal window.

B. Anomaly Detection Approaches

An important task in crop monitoring is the detection of anomalies that can represent risks for the harvest [3], [15]. Detecting nutrient stresses or drought helps to better understand the management of nutrients and, in turn, leads to reduce cultivation costs and increases crop efficiency [2], [16], [17]. Thus, depending on the kind of detected anomalies, the farmers can take action to reduce the adverse effects of the phenomenon that produces the anomaly response.

Anomaly detection (AD) (which includes outlier and novelty detection) is a widely studied problem that relies on the identification of patterns or events that differ from the expected normal behavior of the majority of the data [18]. The existing

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AD methods can be grouped into several categories based on classification/learning, nearest neighbors, clustering, statistical, and deep learning techniques [18], [19] (see [18]–[21] for comprehensive reviews). In particular, AD as a learning task can be supervised, semisupervised, or unsupervised [18], [21], [22], where the major challenge is the lack of labeled training instances [21].

In practical applications, it is generally easier to get access to labeled instances for normal data than getting access to anomalies, which leads to novelty detection [21]. Novelty detection aims at learning a model of normality from a set of data considered as normal to detect unobserved events or novelties in a semisupervised mode [20], [22]. On the other hand, outlier detection is generally defined as the detection of some data points, referred to as outliers, that seem to be inconsistent with the rest of the training set [22]. The main difference between novelties and outliers is that the detected novelties do not always correspond to anomalies [19], [20]. This distinction is interesting in the context of crop monitoring at a parcel level since subtle deviations in data can be the result of external factors such as cloud or forest shadows and not due to anomalies damaging the crops [23], [24]. Several AD algorithms have been investigated in the literature. Some of the most relevant and well-established techniques include the autoencoders (AEs) [25], the local outlier factor [26] or its probabilistic version the local outlier probability [27], the one-class support vector machine (OC-SVM) [28], the isolation forest (IF) [29], and the k-nearest neighbor (kNN) [30].

Some recent works have proposed to solve the AD problem using HMM [31], [32]. In particular, an HMM-based kernel was included in the traditional OC-SVM method in [31]. However, this kernel was defined assuming some specific kind of anomaly, e.g., resulting from mean-value jumps, which is a too strong assumption for crop monitoring. An interesting framework for AD in multivariate time series was proposed in [32], where a set of transformations was used to unify the time series and estimate appropriate features. However, the resulting AD algorithm was trained in a supervised mode, using normal and abnormal labels for the training samples, which are difficult to obtain in most crop monitoring applications.

C. AD for Remote Sensing

In satellite remote sensing images, AD has been conducted either directly on the spectral image pixels [33] or on VIs [15], [34], [35] computed from the combination of several spectral bands. Existing studies such as [15] and [34] exploit historical NDVI data to detect anomalies by comparing new observations against a full set of past observations at the global level. However, analyzing historical data at a parcel level is difficult in the proposed framework because of crop rotation and missing data (due to the presence of clouds that cover some parcels).

Finally, it is interesting to mention some other works such as [10] and [24], which have demonstrated that crops analyzed at a parcel level from multitemporal VIs and vegetation phenology provide suitable knowledge of crops across time. Nevertheless, these studies have not addressed the problem of classifying and identifying possible factors that are damaging the harvest.

D. Proposed Approach

This article introduces a framework for AD and classification of remote sensing time series at a parcel level based on HMM, allowing the detection, temporal localization, and classification of anomalies. The proposed method referred to as anomaly detection based on HMM (AD-HMM) learns the normal dynamic behavior of crops in a given season using several HMM whose parameters are estimated from normal data (i.e., data without any anomaly). Abnormal time series are then detected as those being unlikely to have been generated by these HMM. One advantage of AD-HMM is that the learned HMM can be used for specific time segments of the tested time series, allowing anomalies to be localized during specific time intervals. In a second step, the proposed AD algorithm is complemented by standard classifiers such as SVMs in order to determine the type of detected anomalies.

This article is organized as follows. Section II introduces the proposed framework for detecting, localizing, and classifying anomalies affecting agricultural crops at a parcel level. Section III evaluates the performance of this framework for crop monitoring. Up to our knowledge, this is the first approach providing an AD and classification framework to analyze remote sensing time series at the parcel level for detection, localization, and classification of anomalies, facing limitations such as low-temporal resolution. Consequently, the proposed method cannot be globally compared to a state-of-the-art reference. In order to evaluate the performance of the AD step, we considered some very popular methods such as OC-SVM, IF, and hidden Markov AD (HMAD) in order to appreciate the interest of using HMMs. Numerical experiments conducted using synthetic and real data show the interest of the proposed strategy, allowing abnormal parcels to be detected, localized, and classified.

II. Proposed Method

This section presents the proposed AD and classification approach, which is summarized in the detailed flowchart shown in Fig. 1. Note that the gray shaded squares highlight the main steps of the method: 1) image preprocessing yielding features at the parcel level; 2) learning HMM associated with normal parcels referred to as AD-HMM learning; 3) AD at the parcel level (point AD), which includes the localization of anomalies; and 4) anomaly classification. Sections II-A–II-D describe each procedure following the flowchart of Fig. 1.

A. Image Preprocessing and Feature Extraction

The image preprocessing step requires multitemporal and multispectral images, and the corresponding parcel boundaries (e.g., resulting from a parcellation database such as the land parcel identification system (LPIS)[1] [36]). A set of temporal

[1]LPIS is a system based on images of agricultural parcels used in European countries as a tool to check the eligibility of agricultural land for subsidiary payments.

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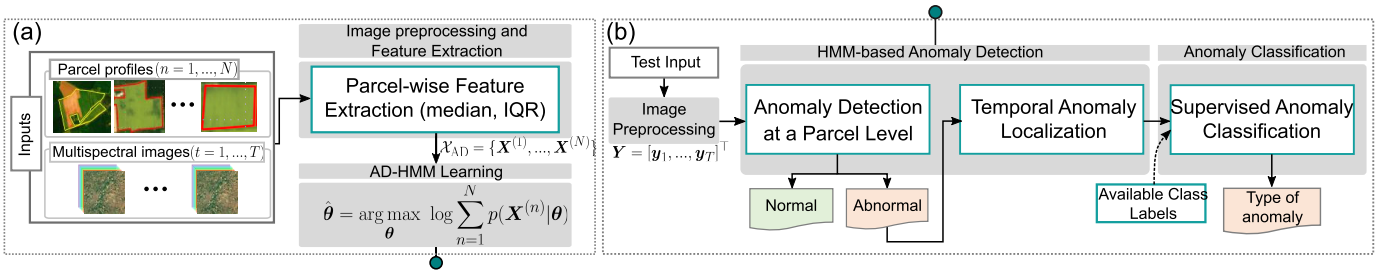


Fig. 1. Flowchart illustrating the main steps and outputs of the proposed approach. (a) Learning step, where multitemporal/multispectral images and parcel profiles are used to extract features of time series for a given parcel. (b) Test step. Gray shaded squares indicate the different tasks, namely image preprocessing, AD-HMM learning, AD, anomaly localization, and anomaly classification.

TABLE I

VIs Estimated From Multispectral Images, Where NIR, R, G, SWIR, and Re Denote the Near-Infrared, Red, Green, Short-Wave Infrared, and Red-Edge Bands

Vegetation Index (VI)	Formula
Normalized difference VI [37], [38]	$\text{NDVI} = \frac{\text{NIR} - \text{R}}{\text{NIR} + \text{R}}$
Green-Red VI [39]	$\text{GRVI} = \frac{\text{G} - \text{R}}{\text{G} + \text{R}}$
Normalized difference water Index (SWIR) [40]	$\text{NDWI}_{SWIR} = \frac{\text{NIR} - \text{SWIR}}{\text{NIR} + \text{SWIR}}$
Normalized difference water Index (Green) [41]	$\text{NDWI}_{G} = \frac{\text{G} - \text{NIR}}{\text{G} + \text{NIR}}$
Modified Chlorophyll Absorption Ratio Index using the Optimized Soil Adjusted VI [42]	$\text{MCARI/OSAVI} = \frac{(\text{Re} - \text{IR}) - 0.2(\text{Re} - \text{R})}{(1 + 0.16)\frac{\text{NIR} - \text{R}}{\text{NIR} + \text{R} + 0.16}}$

TABLE II

Spectral Bands of the Sentinel-2A Multispectral Images Employed in the VI Estimation [24]

Spectral band	Band 3 Green	Band 4 Red	Band 5 Red-Edge	Band 8 NIR	Band 11 SWIR
Wavelength (μm)	0,560	0,665	0,705	0,842	1,610
Resolution (m)	10	10	20	10	20

VIs is extracted from these images. For this study, five VIs derived from the visible, near infrared (NIR), and short-wave infrared (SWIR) were estimated based on images acquired by the Sentinel-2 sensor.[2] These VIs are summarized in Table I and the corresponding spectral bands are shown in Table II.

Two statistical indicators, namely the median and interquartile range (IQR), were computed for each temporal VI, where the IQR is defined by the difference between the 75th and 25th percentiles of the indicator. The motivation for employing statistical indicators for the temporal VIs is that they encompass the mean and dispersion of the VIs with a reduced computational load in the data processing. The preprocessing step provides K features for each parcel of the multitemporal image and for each time instant, i.e., $N \times K \times T$ features, where N is the total number of available parcels and T denotes the number of time instants. These features are the median and the IQR of 5 VIs reported in Table I, leading to $K = 2 \times 5 = 10$ features. To define the extracted time series, let $\boldsymbol{X}^{(n)} = [\boldsymbol{x}_1^{(n)}, \ldots, \boldsymbol{x}_T^{(n)}]^\top$ in $\mathbb{R}^{T \times K}$ denote all the time series computed for the nth parcel (with $n = 1, \ldots, N$), where $\boldsymbol{x}_t^{(n)} \in \mathbb{R}^K$ is the feature vector at time $t \in \{1, \ldots, T\}$ and K is the number of features. Finally, $\mathcal{X}_{\text{AD}} = \{\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(N)}\}$ is the set of N time series extracted from the normal multispectral images and included in the learning set for AD.

The construction of the set $\mathcal{X}_{\text{AD}}$ is referred to as parcel-wise feature extraction since it extracts statistics from the temporal VIs to build a set of features from the multitemporal images and their parcel boundaries. The obtained time series are then validated by experts to make sure that they correspond to normal behavior for AD-HMM learning. Note that the anomalies identified by the expert are excluded from the database and saved (with an anomaly label) in another set denoted as $\mathcal{X}_{\text{AC}}$, which will be used in the anomaly classification procedure (see [24] for further details about the database construction).

B. AD Based on HMM Learning

1) HMM for Temporal VIs: HMMs are doubly stochastic processes defined using an unobservable (hidden) state process, which can be observed via another set of stochastic processes produced by a sequence of observations [9]. HMMs allow the characterization of dynamic systems via a set of hidden states $\boldsymbol{s} = \{s_1, \ldots, s_D\}$ which are inferred from the observations of the system, where D is the number of states in the model. Concisely, an HMM can be formally described by the unknown parameters $\boldsymbol{\theta} = \{\boldsymbol{\pi}, \boldsymbol{A}, \boldsymbol{B}\}$, where $\boldsymbol{\pi} \in \mathbb{R}^D$ is the initial probability vector, which defines the initial probabilities of the system to be in the different states, $\boldsymbol{A} \in \mathbb{R}^{D \times D}$ is the transition probability matrix, which relates the state changes of the hidden latent variable, and $\boldsymbol{B}$ is the emission probability matrix, which is the probability of observing a given value in state s.

In particular, given the nth time series of temporal VIs $\boldsymbol{X}^{(n)}$, the hidden state sequence that reveals a possible state $z_t^{(n)} \in \{s_1, \ldots, s_D\}$ of $\boldsymbol{x}_t^{(n)}$ across time is denoted as $\boldsymbol{Z}^{(n)} = [\boldsymbol{z}_1^{(n)}, \ldots, \boldsymbol{z}_T^{(n)}]^\top$. On the other hand, the entries of the transition probability matrix $\boldsymbol{A}$ are given by $a_{i,j} = P(z_t^{(n)} = s_i | \boldsymbol{z}_{t-1}^{(n)} = s_j)$, which is the probability of transition from a state s_i to the state s_j, for $i, j \in \{1, \ldots, D\}$. Finally, the entries of the emission probability matrix $\boldsymbol{B}$ are given

[2]Sentinel-2 (S2A and S2B) level 2A images were downloaded with a spatial resolution of 10–60 m and a spectral resolution of 13 bands. The theoretical revisit time is five days. Bands with a lower spatial resolution were resampled to obtain a spatial resolution of 10 m × 10 m, and images with a cloud coverage greater than 20% were removed from the database.

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by $b_{i,t} = P(\boldsymbol{x}_t^{(n)}|z_t^{(n)} = s_i)$, which defines the probability density function of the time sample $\boldsymbol{x}_t^{(n)}$ at time t, given that $\boldsymbol{x}_t^{(n)}$ is in the state s_i. More precisely, in the proposed analysis, the emission probability distribution $\boldsymbol{B}$ is assumed to be a mixture of Gaussian distributions, with M multivariate normal densities. Note that the set of states $\boldsymbol{s}$ is typically related to phenological stages that describe the life cycle of vegetation [8], [10], [11].

The detection of anomalies in the AD-based HMM is made using two hypotheses defined as follows:

$$H_0 : \text{Absence of anomaly}$$
$$H_1 : \text{Presence of anomaly}$$

where under hypothesis H_1 a given parcel $\boldsymbol{X}^{(n)}$ is supposed to be abnormal, whereas it corresponds to a normal behavior under hypothesis H_0. The likelihood of a given parcel is defined as

$$P(\boldsymbol{X}^{(n)}|\boldsymbol{\theta}) = \sum_{\text{all } \boldsymbol{Z}^{(n)}} P(\boldsymbol{X}^{(n)}|\boldsymbol{Z}^{(n)}, \boldsymbol{\theta}) P(\boldsymbol{Z}^{(n)}, \boldsymbol{\theta}) = \sum_{z_1^{(n)}, \ldots, z_T^{(n)}} \pi_{z_1^{(n)}} b_{z_1^{(n)},1} a_{z_1^{(n)}, z_2^{(n)}}, \ldots, a_{z_{T-1}^{(n)}, z_T^{(n)}} b_{z_T^{(n)},T}. \quad (1)$$

To correctly model and learn the temporal structure of the underlying data for AD, the HMM model parameter vector $\boldsymbol{\theta}$ is estimated by maximizing the log likelihood, i.e.,

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \log \sum_{n=1}^{N} P(\boldsymbol{X}^{(n)}|\boldsymbol{\theta}) \quad (2)$$

where $\hat{\boldsymbol{\theta}}$ is the parameter vector that better explains $\mathcal{X}_{\text{AD}}$. A local optimal solution of Problem (2) can be found via the Baum–Welch algorithm [9].

2) Generating Different HMM-Models: The estimator $\hat{\boldsymbol{\theta}}$ defined in (2) is associated with all the parcels contained in the training set $\mathcal{X}_{\text{AD}}$. In order to account for different possible structures in the underlying data, this article proposes to build several HMM associated with subsets of N_s samples chosen in $\mathcal{X}_{\text{AD}}$, with $N_s \ll N$. These subsets are built using blocks of time series chosen randomly in $\mathcal{X}_{\text{AD}}$, which leads to L HMM models denoted as $\hat{\boldsymbol{\Theta}} = \{\hat{\boldsymbol{\theta}}_1, \ldots, \hat{\boldsymbol{\theta}}_L\}$, with $\hat{\boldsymbol{\theta}}_\ell = \{\boldsymbol{\pi}^{(\ell)}, \boldsymbol{A}^{(\ell)}, \boldsymbol{B}^{(\ell)}\}$ for $\ell = 1, \ldots, L$. These subsets of time series will be denoted as $\{\mathcal{X}^\ell\}_{\ell=1}^L$ with $\mathcal{X}^\ell \in \mathbb{R}^{N_s \times K \times T}$ and $\mathcal{X}_{\text{AD}} = \bigcup_{\ell=1}^L \mathcal{X}^\ell$. The choice of parameter L will be discussed in Section III.

3) AD-HMM Learning: Algorithm 1 summarizes the main steps of the proposed AD-HMM learning, corresponding to the second gray box in Fig. 1(a). This algorithm receives the set of time series $\mathcal{X}_{\text{AD}}$, the number N_s of subsets used to learn a single model, the number of HMM states D, and the number of models to be learned L. Default values resulting from simulations conducted on real images are provided for each parameter. In the initialization step (Line 1), the number of components M used in the Gaussian mixture model used for the emission probability distribution introduced in Section II-B1 was computed directly from the input data. A strategy to estimate this number of Gaussians is to use the number of local maxima in the data histogram and to reduce this number until the algorithm performance decreases significantly. In the next step, a random index selection of N_s time series stacked in $\mathcal{X}^\ell$ is performed, indicating the parcels to be selected from the set $\mathcal{X}_{\text{AD}}$, as detailed in Section II-B2. The HMM model parameters $\boldsymbol{\theta}$ are then randomly initialized [9]. Finally, the HMM model parameters are estimated using the Baum–Welch procedure and stacked into the set $\hat{\boldsymbol{\Theta}} = \{\hat{\boldsymbol{\theta}}_1, \ldots, \hat{\boldsymbol{\theta}}_L\}$, where $\hat{\boldsymbol{\theta}}_\ell = \{\boldsymbol{\pi}^{(\ell)}, \mathbf{A}^{(\ell)}, \mathbf{B}^{(\ell)}\}$.

Algorithm 1: AD-HMM Learning Procedure

Input: $\mathcal{X}_{\text{AD}}$: Set of time series associated with parcels;
N_s: # of images per model (default $N_s = 100$);
L: # of models to be learned (default $L = 10$);
D : # of states (default $D = 18$);

1 **Initialize:** M: Estimate the number of Gaussian mixtures;
2 **for** $\ell = 1$ **to** L **do**
3 Built $\mathcal{X}^\ell$ by randomly selecting N_s parcels in $\mathcal{X}_{\text{AD}}$;
4 Initialize $\boldsymbol{\theta}_\ell = \{\boldsymbol{\pi}^{(\ell)}, \boldsymbol{A}^{(\ell)}, \boldsymbol{B}^{(\ell)}\}$;
5 $\hat{\boldsymbol{\theta}}_\ell \leftarrow$ BAUM-WELCH($\mathcal{X}^\ell, \boldsymbol{\theta}_\ell, M, D$);

Output: L HMM models $\hat{\boldsymbol{\Theta}} = \{\hat{\boldsymbol{\theta}}_1, \ldots, \hat{\boldsymbol{\theta}}_L\}$ and their subsets of time series $\mathcal{X}^\ell$.

C. HMM-Based AD

For the testing part [see Fig. 1(b)], AD is first performed at the parcel level to detect abnormal parcels. The detected anomalies are then localized in time as explained in Section II-C2. The proposed strategy is composed of a point AD step detecting abnormal parcels and a contextual AD [18] step allowing the starting time of the anomaly to be estimated.

1) AD at the Parcel Level: The probability that a time series $\boldsymbol{Y} = [\boldsymbol{y}_1, \ldots, \boldsymbol{y}_T]^\top$ has been generated by the ℓth learned model is written as

$$w_\ell = \log P(\boldsymbol{Y}|\hat{\boldsymbol{\theta}}_\ell) \quad (3)$$

for $\ell = 1, \ldots, L$, where $\boldsymbol{w} = [w_1, \ldots, w_L]^\top \in \mathbb{R}^L$ is a vector containing the log probabilities of the test signal with respect to the L HMM models learned using Algorithm 1. Note that the log probability in (3) is determined following the procedure in Table III, i.e., using the forward algorithm [9], where $\alpha_{i,t}$ is the probability of partial observation of the time series at time t and state i (so-called forward variable) and $a_{i,j}^{(\ell)}$ and $b_{j,t}^{(\ell)}$ are the elements of the matrices $\boldsymbol{A}^{(\ell)}$ and $\boldsymbol{B}^{(\ell)}$ for the ℓth model with parameter vector $\hat{\boldsymbol{\theta}}_\ell$, respectively. Consequently, (3) is the sum of the forward variables $\alpha_{i,t}$ across t providing a unique probability w_ℓ for a given parcel.

The final AD rule (at the parcel level) is defined as

$$w^* = \max_{\ell=1,\ldots,L} w_\ell \underset{H_0}{\overset{H_1}{\lessgtr}} \tau \quad (4)$$

where w^* is the probability associated with the most likely model class (with the highest probability) and τ is a threshold related to the probability of false alarm (PFA) and the probability of detection (PD) of the test. More precisely, the value of τ was determined as the point of the ROC curve (expressing the PD as a function of the PFA) located closest to the ideal point

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TABLE III
ESTIMATION OF LOG PROBABILITIES FOR A TEST SIGNAL

Forward-procedure for the $\ell = 1, ..., L$ models	
1) $\alpha_{i,1} = \pi_i^{(\ell)} b_{i,t}^{(\ell)}$	(Initialization)
2) $\alpha_{i,t+1} = \left[\sum_{j=1}^{D} \alpha_t \; a_{i,j}^{(\ell)}\right] b_{j,t}^{(\ell)}$	(Induction)
3) $\log P(\boldsymbol{Y}\|\hat{\boldsymbol{\theta}}_\ell) = \sum_{j=1}^{D} \alpha_{i,T}$	(Ending)

(PFA, PD) = (0, 1) [43]. Looking carefully at the proposed AD rule (4), a tested time series is declared as normal if the highest probability w^* exceeds the threshold τ, i.e., if a least one of the L models associated with the normal HMM models is likely to correspond to the observations. This detection rule is motivated by the fact that it is assumed that L different HMM models capture the possible temporal structures or signal dynamics of normal time series.

2) Anomaly Localization via Segmentation: When a tested time series $\boldsymbol{Y}$ has been declared as abnormal in (4), it goes into the second step devoted to anomaly localization. In this step, the HMM models $\hat{\boldsymbol{\Theta}} = \{\hat{\boldsymbol{\theta}}_1, \ldots, \hat{\boldsymbol{\theta}}_L\}$ determined using Algorithm 1 are used on time segments $\left[t_{\rho_1}, t_{\rho_2}\right] = \{t \,|\, t_{\rho_1} \leq t \leq t_{\rho_2}\}$ (instead of analyzing the complete time series) to determine the starting point of the anomaly in the time series. Consider the forward variable $\alpha_{i,t-1}$ at time $t-1$ in its scaled version defined as $\tilde{\alpha}_{i,t-1} = \alpha_{i,t-1} / \sum_{i=1}^{D} \alpha_{i,t-1}$, where $i = 1, \ldots, D$, with D the number of HMM states. The probability of having $\boldsymbol{Y}$ generated by the model $\hat{\boldsymbol{\theta}}$ at time t can be written it terms of $\alpha_{i,t-1}$ as follows:

$$u_t = 1 \Big/ \left(\sum_{i=1}^{D} \tilde{\alpha}_{i,t-1} a_{i,j} b_{i,t-1} \right) \tag{5}$$

where u_t depends on the scaled forward variable $\tilde{\alpha}_{i,t-1}$, the transition probability $a_{i,j}$, and the emission probability at time $t-1$. Note that this expression for u_t results from the first-order Markov chain rule, which assumes that the current state (at time t) depends only on its predecessor state (at time $t-1$) [9]. Note also that the forward variable $\alpha_{i,t-1}$ is used in its scaled version $\tilde{\alpha}_{i,t-1}$ to avoid overflow. Indeed, this variable relies on the sum of a large number of terms, as shown in the induction step of the forward procedure. The log likelihood of the time series in the time segment $[t_{\rho_1}, t_{\rho_2}]$ is defined as

$$\log P(\boldsymbol{y}_{t_{\rho_1}}, \ldots, \boldsymbol{y}_{t_{\rho_2}} | \boldsymbol{\pi}, \boldsymbol{A}, b_{i,[t_{\rho_1}-1, t_{\rho_2}-1]}) = -\sum_{t=t_{\rho_1}}^{t_{\rho_2}} u_t \tag{6}$$

where u_t has been defined in (5) and $b_{i,[t_{\rho_1}-1, t_{\rho_2}-1]}$ is the emission probability for the time interval $\left[t_{\rho_1}, t_{\rho_2}\right]$, for $i = 1, \ldots, D$. As a result, the probability that a time series $[\boldsymbol{y}_{t_{\rho_1}}, \ldots, \boldsymbol{y}_{t_{\rho_2}}]$ has been generated by the ℓth learned model on the segment $[t_{\rho_1}, t_{\rho_2}]$ can be computed as

$$\boldsymbol{w}_{\ell,[t_{\rho_1}, t_{\rho_2}]} = \log P\left(\boldsymbol{y}_{t_{\rho_1}}, \ldots, \boldsymbol{y}_{t_{\rho_2}} | \boldsymbol{\pi}^{(\ell)}, \boldsymbol{A}^{(\ell)}, b_{i,[t_{\rho_1}-1, t_{\rho_2}-1]}^{(\ell)}\right) \tag{7}$$

where $\boldsymbol{w}_{\ell,[t_{\rho_1}, t_{\rho_2}]}$ is the vector containing the log probabilities of the different models in the time interval $[t_{\rho_1}, t_{\rho_2}]$.

The anomaly localization studied in this article considers four time intervals associated with different phenological stages of crop growth to identify potential anomalies. These intervals are referred to as *Growing*, *Flowering*, *Adult-phase*, and *Senescence* [shown in Fig. 8(a)], as in [11] and [13]. The hypothesis test to localize the anomaly in a given interval is defined as

$$w_S = \max_{\ell=1,\ldots,L} w_{\ell,[t_{\rho_i}, t_{\rho_j}]} \underset{H_0}{\overset{H_1}{\lessgtr}} \tau_S \tag{8}$$

where $S \in \{$*Growing, Flowering, Adult-phase, Senescence*$\}$ denotes the growth stage under analysis in the hypothesis test, τ_S is a threshold depending on the PFA and the PD of the test sample for a given stage S, t_{ρ_i} and t_{ρ_j} are the beginning and the end of the stage, respectively, and w_S contains the probabilities the phenological stage S given the tested time series. Note that (4) defines the parcel detection rule, whereas (8) localizes the anomaly in one of the predefined phenological stages.

D. Anomaly Classification

A final step can be included in the analysis to identify the anomaly that has affected the parcel $\boldsymbol{Y}$ using a supervised classifier. After an anomaly has been localized using the steps shown in Fig. 1(b), we propose to classify the detected anomaly into one of the C classes defined by the user and corresponding to the possible types of anomalies affecting the analyzed crop.

The set of features used for the classification is composed of the abnormal time series that has been detected in (4), whose different feature vectors are introduced as columns in the input matrix. More precisely, the feature matrix for training the classifier is of the form $\boldsymbol{X}_{\text{AC}} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_R]$, where $\boldsymbol{x}_r = [x_{r,1}, \ldots, x_{r,KT}]^\top$ is a vector containing the KT features extracted from the rth time series at all time instants. It is important to highlight that each time series $\boldsymbol{x}_r$ selected for training the classifier contains an abnormal time series from $\mathcal{X}_{\text{AC}}$ with the corresponding label, denoted by $\boldsymbol{v}_r \in \{1, \ldots, C\}$, where C is the total number of classes. In the testing part, the classifier generates for each time series $\boldsymbol{y} = [y_1, \ldots, y_{KT}]^\top$ a label $v_{\boldsymbol{y}} \in \{1, \ldots, C\}$ indicating the class of the anomaly of $\boldsymbol{y}$.

III. RESULTS

The performance of the proposed methodology[3] is evaluated on both simulated and real data. AD is evaluated in terms of precision, recall, and area under the precision-vs-recall curve (AUC), whereas the overall accuracy (OA), the kappa coefficient, and the probabilities of correct classification are used for anomaly classification.[4] The precision and recall are defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

where TP, TN, FP, and FN are the numbers of true positives, true negatives, false positives, and false negatives,

[3]For the AD-HMM implementation, we used the HMM toolbox available online at https://www.cs.ubc.ca/~murphyk/Software/HMM/hmm.html

[4]The higher the value of the metric, the better the detection or classification.

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respectively.[5] On the other hand, the probability of correct classification for class c (denoted as P_c) is defined as

$$P_c = \frac{1}{R_c} \sum_{r=1}^{R_c} \delta(v_r, \hat{v}_r) \tag{9}$$

where R_c is the total number of training vectors of the class $c \in \{1, \ldots, C\}$, v_r and $\hat{v}_r$ are the true and estimated labels of the rth training vector of class c, respectively, and $\delta(\cdot)$ is an indicator such that $\delta(v_r, \hat{v}_r) = 1$, if $v_r = \hat{v}_r$, and zero otherwise.

A. Synthetic Data

Synthetic data are first considered to evaluate the AD-HMM approach for point and contextual AD. For point AD, the proposed approach is compared against three state-of-the-art algorithms: the OC-SVM with a Gaussian [or radial basis function (RBF)] kernel, the IF, and the HMAD [31] with a linear kernel. The OC-SVM and IF algorithms have been chosen based on their high impact on AD. HMAD was considered since it is a recent AD method using HMM differently. For a fair comparison with the proposed approach that relies on training and testing phases, the OC-SVM and IF are used in a semisupervised mode (or novelty detection mode), denoted as OC-SVM-N and IF-N, to conduct both training and testing steps. The best values of the parameters for each algorithm were selected, leading to 1000 isolation trees and a subsampling ratio of 256 for IF, an outlier ratio of $\nu = 0.1$ for HMAD, and an outlier fraction of $\nu = 0.1$ for OC-SVM-N. Note that the value of the kernel bandwidth γ used in the RBF kernel was estimated using the median of the pairwise Euclidean distances between vectors of the training feature set, as suggested in [44]. The number of Gaussians of the mixture model (denoted as M) was estimated using the synthetic data, leading to values of M varying from 20 to 55.

In the first experiment, synthetic time series were generated as zero-mean Gaussian sequences with unit variance for the normal data, i.e., with mean $\mu = 0$ and variance $\sigma^2 = 1$. Mean-value jumps (first scenario) and variance changes (second scenario) were then introduced as anomalies in the testing sequences to evaluate two anomaly scenarios. Specifically, the number of normal time series generated for the learning phase was $N = 1000$ with $T = 300$ temporal samples. For the testing phase, a set of 500 sequences was generated, with 10%, 20%, 30%, and 50% of abnormal time series, where anomalies are introduced in the time series as blocks of 90 contiguous samples.

Fig. 2 compares the different AD algorithms in terms of average AUC (the AUC values were obtained from ten Monte Carlo runs for each probability of anomalies). This figure shows that the AUC values when the percentage of anomalies is 30% exhibit an inflection point, where the curve changes from a relative low-to-high and steady value in mean-value jumps, and a relative high-to-low AUC value in the variance changes plot. To complement these results, Fig. 3

[5]The detection threshold was determined using the point of the AUC curve located the closest to the ideal point (0, 1).

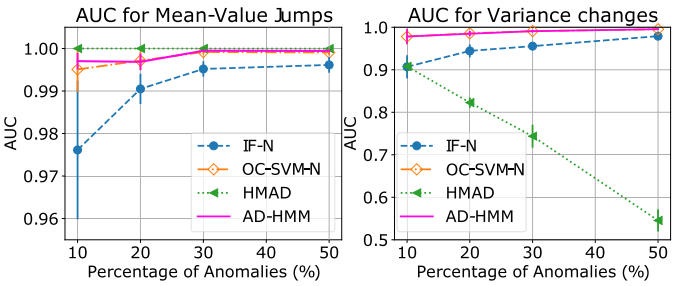


Fig. 2. AD performance in terms of AUC for different percentages of anomalies in the testing set, where the proposed AD-HMM and the OC-SVM-N methods obtain the overall best performances.

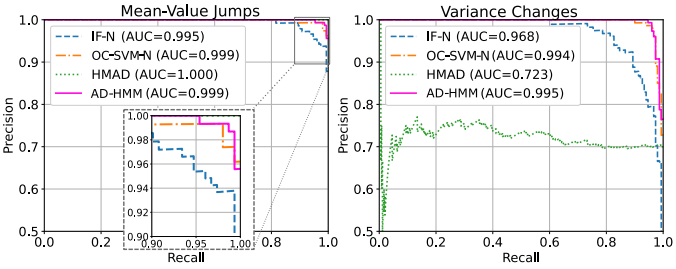


Fig. 3. Illustration of the precision-vs-recall curves for each AD algorithm when the percentage of mean-value jumps and variance changes anomalies in the testing set is 30%.

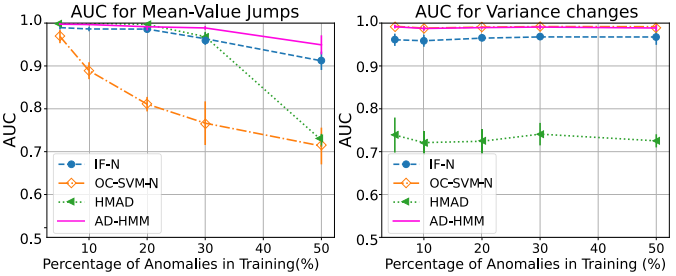


Fig. 4. AD performance in terms of AUC for different percentages of anomalies in the training set when the testing set has 30% of anomalies. Note that the proposed AD-HMM provides the overall best detection.

shows the examples of AUC curves obtained when there are 30% of anomalies in the testing set. In general, the proposed AD-HMM reaches comparable results to those obtained with the OC-SVM-N algorithm, with the advantage of being able to localize the anomalies (see details below). In the case of variance changes, Fig. 2 shows that the HMAD performance decreases for large percentage of anomalies, reflecting the fact that this algorithm was designed for anomalies characterized by mean-value jumps.

An additional controlled experiment was conducted to evaluate the robustness of the AD algorithms to possible anomalies present in the training set, with percentages of anomalies of 5%, 10%, 20%, 30%, and 50%. The results displayed in Fig. 4 show that the proposed method seems to be more robust to the presence of anomalies in the training set, when compared to the state of the art.

In order to evaluate the performance of the proposed anomaly localization (contextual AD) on data with controlled

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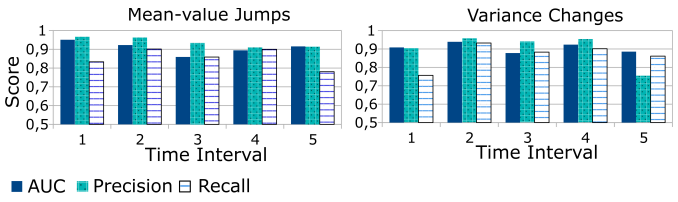


Fig. 5. Summary of the time anomaly localization (contextual AD) performance on each synthetic time interval for the proposed AD-HMM method, where, in general, the obtained AUC values per time interval (dark blue bars) are higher than 0.85.

ground truth, we simulated anomaly time intervals by considering time segments of 60 time samples, leading to five intervals. The time series detected as abnormal after the proposed AD-HMM approach were then analyzed to localize the detected anomalies. Note that the synthetic anomalies were generated according to the two scenarios corresponding to mean-value jumps with amplitude $\mu = 1.2$ and variance changes with a ratio of variances equal to 1.5. These scenarios correspond to examples of typical anomalies that can affect crop parcels due to growing or heterogeneity problems. Fig. 5 summarizes the localization performance on each time interval from the synthetic dataset for mean-value jumps and variance changes, where each bar of the plot corresponds to the AUC, precision, and recall.

As a complement, Fig. 6 shows some anomaly localization results obtained for time series subjected to anomalies defined by mean-value jumps (top) or variance changes (bottom). Note that the abnormal time series are represented using a solid line and that the $\star$ labels indicate the samples affected by the anomaly (ground truth). The time intervals considered by the algorithm are located between the dotted vertical lines and those detected as abnormal using the algorithm are shown using the shaded orange boxes. These results illustrate the anomaly localization performance of the proposed method.

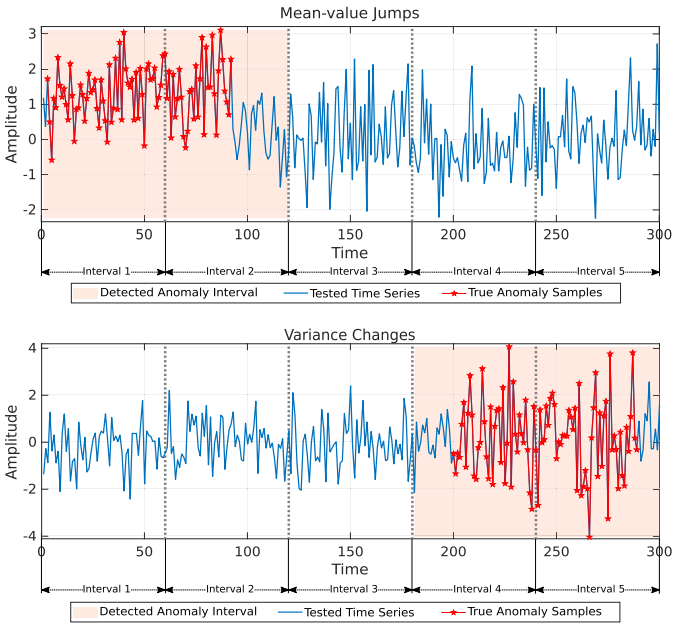


Fig. 6. Examples of anomaly localization for synthetic data subjected to (Top) mean-value jumps and (Bottom) variance changes. The solid line represents the tested time series and the $\star$ line indicates the time samples affected by the anomaly. The time intervals considered in the analysis are delimited by the gray dotted vertical lines and the segments detected as abnormal are highlighted with a shaded orange box.

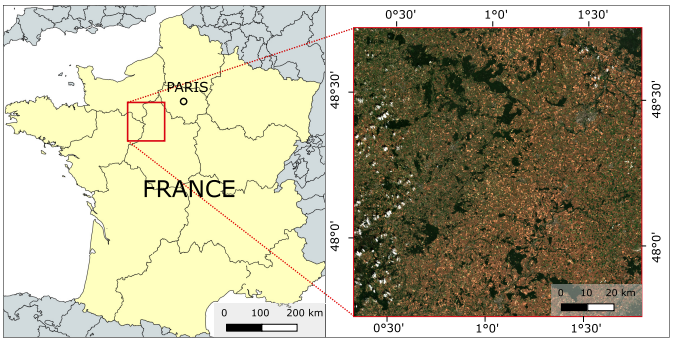


Fig. 7. Study area located in Beauce, North of France.

B. Experiments on Real Data: Multispectral Rapeseed Images

1) Study Area: The research site displayed in Fig. 7 is located in Beauce, North of France. This site contains a lot of crop fields, such as rapeseed and wheat. A set of 13 multispectral Sentinel-2 images was selected between October 2017 and June 2018. The dataset was processed to level 2A using the MAJA[6] processing chain [45]. A set of 1924 rapeseed parcels was extracted from the images with the support of agronomists, as shown in Fig. 1(a). The resulting dataset was analyzed with the aim of labeling part of the data for evaluation purposes. The anomalies found in the data were related to vegetation phenology problems, crop heterogeneity, boundary errors, wrong crop type, and shadow perturbations. Vegetation phenology problems [shown in Fig. 8(a)] are associated with problems during plant development, such as early/late growth, early/late senescence (plant degradation), and early flowering.

[6]MAJA is available on the Plateforme d'Exploitation des Produits Sentinel (PEPS) platform of the French National Center for Space Studies [Centre National d'Études Spatiales (CNES)].

On the other hand, heterogeneity corresponds to the spatial heterogeneous development of the crop, which can indicate the presence of crop or soil diseases and can affect the crop in any moment of growth. Boundary errors and wrong crop type are anomalies describing problems related to the database delineation and wrong target crop. Note that the rapeseed data and their corresponding labels are the same as those used in [24]. It is important to note that the labeling used for the real data was performed with the help of agronomic experts by visual interpretation using all the available images (where each abnormal parcel was manually verified) and by using all the time series of the different VI features to compare any analyzed parcel to the rest of the dataset using outlier scores resulting from unsupervised outlier detection algorithms [24]. Finally, observe in Fig. 8(a) that even though there are four images/samples per growing stage, the growth problems were identified by the experts with very high confidence.

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The partitioning of the dataset for AD was randomly performed as follows: $N = 500$ normal parcels were selected for AD-HMM learning and the remaining 1424 parcels (697 normal parcels and 727 anomalies) were considered for the testing phase. The test parcels detected as anomalies were then considered as input data for the classifiers. Based on the anomalies found in the database, the anomaly classification step considers the following classes: late growth, early senescence, late senescence, and other, where "other" is a class containing all the anomalies that do not belong to the previous three classes. These anomalies are nonagronomic anomalies (such as errors in parcel boundaries, wrong crop type, or shadow perturbations) or agronomic anomalies affecting a very small number of parcels (such as early growth, early flowering, and crop heterogeneity).

For illustration purposes, Fig. 8(a) and (b) shows the expected temporal profile of the NDVI median with anomalies related to problems in the vegetation phenology such as early/late growth, early/late senescence, and early flowering, and the distribution of a set of 500 normal and abnormal NDVI medians, respectively. As can be seen in the histograms and the zoom, the normal and abnormal data have different distributions, allowing anomalies to be classified. Note that the different vegetation phenological stages for the rapeseed crops indicated in the top of Fig. 8(a) are located between the vertical gray dotted lines and were selected based on [2] and [24].

2) Analysis of VIs: To analyze the impact of using different time series of VIs for AD, Fig. 9 compares the performance of the proposed AD-HMM algorithm using the median and IQR of different combinations of VIs introduced in Table I. More precisely, Fig. 9 shows the AUC values obtained for different VI combinations after averaging the results of ten Monte Carlo runs. Note that all the VIs were scaled such that each column of the feature matrix takes its values in the interval (0, 1) (minimum–maximum scaling). One can observe that the performance of AD-HMM is very similar when using different VI combinations. Therefore, the remaining analyses will be performed with NDVI only. When using NDVI only, the estimated number of Gaussians in the mixture model was fixed to $M = 6$.

The overall performance of the proposed AD-HMM algorithm depends on the parameters N_s, L, and D that need to be adjusted. The number of states was varied in the set $\{3, 9, 12, 15, 18\}$, whereas the values of N_s and L were chosen in the set $\{10, 25, 50, 100, 200\}$. For $D = 18$, the averaged AUC metrics vary in the interval $[0.80, 0.83]$, and the best performance for all the VIs was obtained when $N_s = 100$ and $L = 10$. These values were selected in the rest of the analysis, in particular to display Fig. 9.

3) AD Results: The experiments conducted on real data using the proposed AD approach were compared to different algorithms, including IF-N, OC-SVM-N, and HMAD. All the algorithms were run using the NDVI features. The parameters of the algorithms were set by cross validation. For OC-SVM-N, the outlier ratio was set to $\nu = 0.1$, and the kernel parameter for the RBF kernel was estimated using Jaakkola's heuristic [44]. The IF-N algorithm was run using

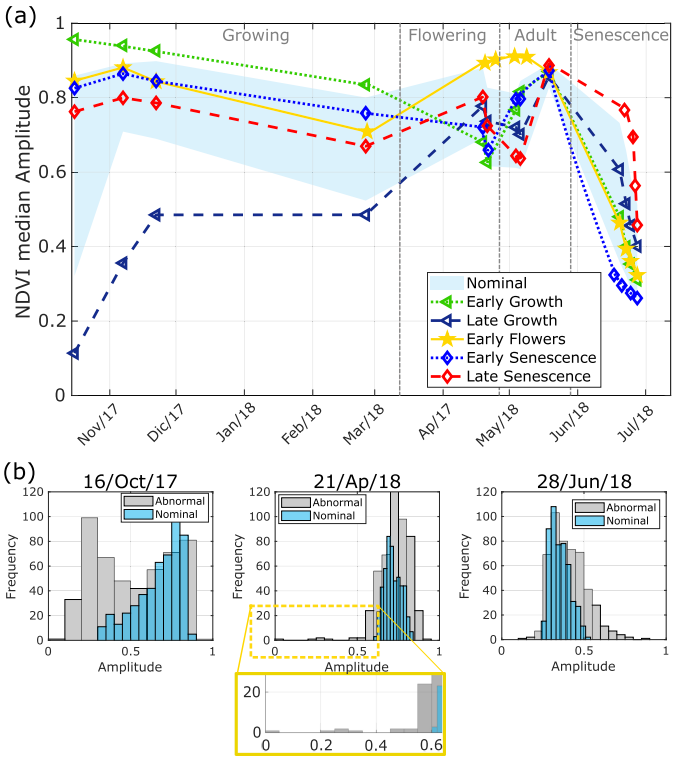


Fig. 8. Temporal profiles and distribution of normal (blue) and abnormal data. (a) Five typical time series profiles for agronomic anomalies are shown, where the shaded blue section corresponds to the normal time series. (b) Histogram of 500 time series of normal (blue) and abnormal (gray) NDVI median for three dates, which illustrates how the distribution of abnormal data deviates with respect to the normal data, leading potential anomalies to be detected by the proposed approach.

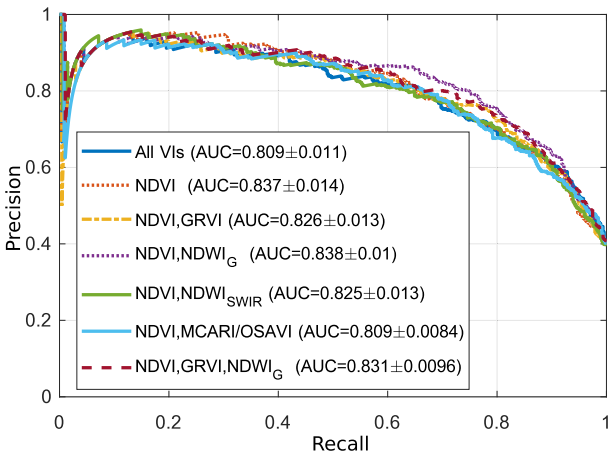


Fig. 9. Performance of the AD-HMM detection using the median and IQR of different temporal VIs, where the AUC value of each VI combination is shown in the legend.

1000 isolation trees and a subsampling ratio of 256. The outlier fraction used in HMAD was set to $\nu = 0.3$. Fig. 10 summarizes the performance of the different algorithms, where the proposed approach obtains slightly better results than OC-SVM-N and IF-N. The poor performance obtained with HMAD is probably due to the fact that the anomalies affecting crop parcels are not limited to mean changes but also lead

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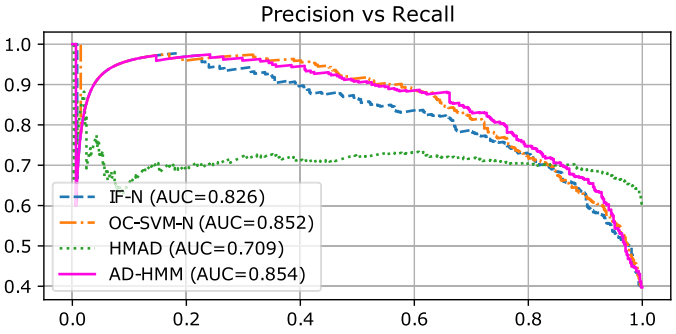


Fig. 10. Performance of different AD methods to detect abnormal parcels in the real dataset.

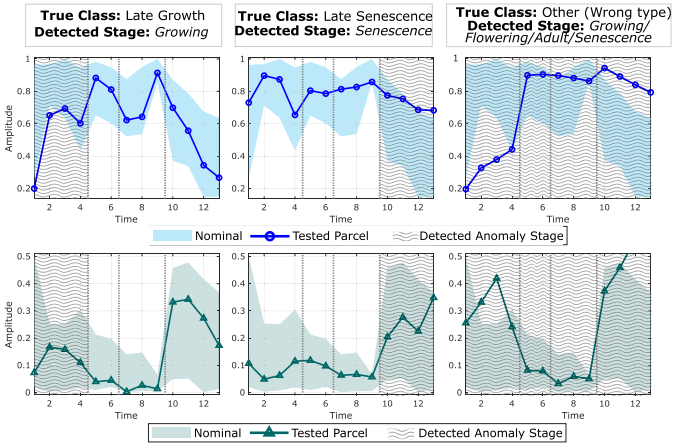


Fig. 11. Time anomaly localization for three tested parcels of rapeseed crops affected by different anomalies. Each plot displays (Top) median and (Bottom) IQR of the NDVI features. The box in the top indicates the class of the anomaly and the detected stage. The lattice box highlights the detected stage.

to variance changes for which the HMAD algorithm is not adapted, as shown in the synthetic results for variance changes [see Fig. 2 (right)].

4) Time Anomaly Localization: The signals detected as abnormal in the previous step were then analyzed to localize the anomalies affecting the crops. For an easier interpretation, the acquisition dates presented in Fig. 8 were transformed into integer values following the time intervals associated with the different phenological stages of rapeseed crops as follows: Growing $= \{t|1 \leq t \leq 4\}$, Flowering $= \{t|5 \leq t \leq 6\}$, Adult $= \{t|7 \leq t \leq 9\}$, and Senescence $= \{t|10 \leq t \leq 13\}$.

Fig. 11 shows the results obtained for three time series with growth and wrong-type problems, where the estimated value of $S \in \{Growing, Flowering, Adult\text{-}phase, Senescence\}$ by the proposed AD-HMM is indicated in the top of each figure as "Detected Stage," whereas the class of the anomaly is referred to as "True Class." The lattice box on the plots in Fig. 11 highlights the detected stage. Note that based on the learned models, the proposed approach can estimate when the temporal structure deviates from the normal behavior, even, for subtle deviations as shown in the plot of the middle for *Late Senescence* problems.

TABLE IV

PERFORMANCE RESULTS FOR THE DIFFERENT CLASSIFIERS (LOOC)

#	Classes	# Samples	SVM-LN	SVM-RBF	KNN	RF
1	Late Growth	179,1 (5,8)	0,79 (0,01)	**0,82** (0,00)	0,81 (0,01)	0,81 (0,01)
2	Early Sen.	51,3 (3,0)	0,91 (0,03)	**0,94** (0,02)	0,84 (0,04)	0,72 (0,02)
3	Late Sen.	28,8 (1,0)	0,97 (0,01)	**1,00** (0,01)	0,92 (0,01)	0,73 (0,03)
4	Other	352,3 (1,8)	0,59 (0,01)	0,60 (0,00)	0,61 (0,01)	**0,76** (0,01)
Average Pc			0,81 (0,01)	**0,84** (0,00)	0,79 (0,01)	0,75 (0,01)
OA (%)			69,16 (0,42)	71,32 (0,43)	70,26 (0,90)	**77,04** (0,49)
kappa			0,54 (0,01)	0,57 (0,01)	0,54 (0,01)	**0,62** (0,01)

TABLE V

CONFUSION MATRIX FOR THE SVM-RBF CLASSIFIER FOR REALIZATION # 10 (PC: 0.83, OA: 70.1%, AND KAPPA: 0.57)

Detected AD-HMM: **618**		Predicted Class			
Classes		Late Growth	Early Senescence	Late Senescence	Other
True Class	Late Growth	**151**	6	16	11
	Early Senescence	1	**49**	0	4
	Late Senescence	0	0	**28**	0
	Other	55	61	26	**210**

C. Anomaly Classification After AD and Localization

The last step of the proposed algorithm classifies some classes of anomalies detected in the rapeseed crops. To evaluate the classification performance on the available samples, the leave-one-out cross-validation (LOOC) strategy was considered. LOOC consists in leaving one vector out of the database, training the classifier with all the remaining samples, testing the classifier with the vector removed from the database, and repeating these operations R times, where R is the size of the database. This strategy was selected given the few number of training samples available for anomaly classification. The classifiers considered in this section were the random forest (RF) algorithm with 100 trees and a maximum number of features set to the square root of the number of columns of the feature matrix, the kNN classifier with $k = 3$, and the support vector machine algorithm with linear (SVM-LN) and Gaussian (SVM-RBF) kernels, with a regularization parameter $C = 1$. The different parameters values of the classifiers were chosen in order to obtain the best performance. The multiclass strategy used in the SVM-based classifiers was based on the one-against-one voting strategy [46]. In addition, the synthetic minority oversampling technique (SMOTE) was used to oversample the training set to mitigate the unbalanced nature of the dataset [47].

Table IV shows the estimated probability of correct classification, OA, and kappa coefficient for the different classifiers obtained using ten Monte Carlo runs (where the highest values are highlighted in bold and the corresponding standard deviations are indicated in brackets). Note that the number of samples is presented in average. As can be observed, the highest value of P_c is obtained for the SVM-RBF classifier, whereas the better OA and value of kappa were obtained from the RF classifier. These two classifiers provide the overall best classification performance. The resulting confusion matrix of the SVM-RBF classifier for the tenth realization is shown

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in Table V, where 618 samples were detected using AD-HMM. Note that the class *other*, which contains anomalies such as heterogeneity, wrong crop type, errors in parcel boundaries, and shadow perturbations, allows us to be close to a real scenario where anomalies that cannot be explained by abnormal plant growing are often present. It is important to mention here that in the rapeseed crops of this study, those classes (wrong crop type, heterogeneity, and shadow perturbations) affect either the whole time series or some time intervals in a random way, yielding anomalies located in any time interval. This lack of structured patterns increases the complexity of the classification, which explains the relatively poor classification performance obtained for this class. Additional information resulting from other data, e.g., from synthetic aperture radar images, might be considered to improve the classification performance. This work is currently under investigation.

IV. Conclusion

This article studied a method for detecting, localizing, and classifying anomalies that affect agricultural crops based on HMMs and machine learning. The proposed AD-HMM exploited the temporal structure of time series of VIs extracted from multispectral images to perform both point (parcel-wise) and contextual (temporal-wise) anomaly detection. The proposed method also allowed the detected anomalies to be temporally localized and classified into predefined classes, information that is valuable for crop monitoring. A comparison with classical AD algorithms, in terms of precision and recall, provided very promising results. An interesting property of the proposed AD algorithm is its capacity of localizing and classifying the anomalies located within each time series by exploiting the previously learned HMM models, where the outcomes of these steps provide complementary knowledge to the farmers and producers for monitoring their crops.

Further investigation should be conducted to evaluate the interest of the proposed approach for detecting anomalies in other kinds of crops to characterize their dynamic behavior. Moreover, it would be interesting to generalize the proposed approach to nonhomogeneous Markov chains in order to handle transitions between states defined by nonstationary time series. Another interesting further work is the extension of the proposed AD-HMM to time series of VIs estimated from multiple remote sensing sources, e.g., extracted from synthetic aperture radar (SAR) images or vegetation optical depth (VOD) retrievals derived from microwave sensors (these VIs have been used in phenology studies in [4]). Finally, it could also be interesting to investigate the application of the proposed approach to features estimated from other kinds of sensors such as compressive multitemporal/multispectral sensors [48], [49], which acquire the images using a compressed format.

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